



Article

A Combined Deep Learning Approach for the Early Detection of Respiratory Disease Using Multiple Environmental Sensors

Jake Cowton ^{1,2,*}, Ilias Kyriazakis ² , Thomas Plösch ³

¹ Interdisciplinary Computing and Complex BioSystems, Newcastle University, Newcastle upon Tyne NE1 7RU, UK

² Agriculture, School of Natural and Environmental Sciences, Newcastle University, Newcastle upon Tyne NE1 7RU, UK; ilias.kyriazakis@newcastle.ac.uk

³ Open Lab, Newcastle University, Newcastle upon Tyne NE1 7RU, UK

* Correspondence: j.cowton2@newcastle.ac.uk

† Current address: School of Interactive Computing, Georgia Institute of Technology, Atlanta, GA 30332, USA

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Abstract: We designed and evaluated an assumption-free approach for the early detection of animal health monitoring, specifically for the early detection of respiratory disease based on environmental sensor data. Two recurrent neural networks (RNNs), consisting of two recurrent units (GRUs), were used to create an autoencoder. Environmental data, collected from a variety of sensors, was processed by the autoencoder. A prototype of network trained to reconstruct the patterns of environmental data that did not lead to an occurrence of respiratory disease. A pattern of “healthy environmental data” had a greater likelihood of being reconstructed. Data were labelled as either normal or anomalous using particle swarm optimisation (PSO), from which a model was developed. This method outperformed state-of-the-art techniques, resulting in a more accurate detection of the actual observations. The results show that a change in the behaviour of pigs showing symptoms of respiratory disease can be detected at an early time during which their keepers can act to mitigate the spread of the disease. Porcine reproductive and respiratory syndrome (PRRS) is a major concern in pigs.

Keywords: anomaly detection; deep learning; sensors; pig health; PRRS

1. Introduction

The forecasting of an oncoming health challenge is a major concern for the level of health and animal welfare. The earlier a disease is detected, thus lowering the overall impact of the disease on the animal, the greater the likelihood of treatment success [1]. Respiratory disease, such as porcine reproductive and respiratory syndrome (PRRS), pneumonia, and pleurisy, are so common in commercial pig populations [2]. Therefore, it is important to detect the onset of respiratory disease, which we define as the point at which the disease increases from 0 to greater than 0.

There are many factors that influence the conditions of the environment which the pigs inhabit. A number of studies have been conducted—specifically, air quality, temperature, and humidity—by Stark et al. [2].

Using GRU-Autoencoder for Respiratory Disease in Pigs with Environmental Sensors

Thomas Plötz ^{3,†}  and Jaume Bacardit ¹ 

¹ Systems (ICOS) Research Group, School of Computing,
Newcastle University, Newcastle upon Tyne NE1 7RU, UK; jaume.bacardit@newcastle.ac.uk

² School of Natural Sciences, Newcastle University,

Newcastle upon Tyne NE1 7RU, UK; thomas.ploetz@gatech.edu

³ School of Computer Science, Georgia Institute of Technology, Atlanta, GA 30318, USA.

† Correspondence: thomas.ploetz@gatech.edu

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This paper presents a new, assumption-free, deep learning-based methodology for early detection of respiratory disease in growing pigs. The methodology consists of recurrent neural networks (RNNs), each comprising a gated recurrent unit (GRU) and an autoencoder (GRU-AE) into which environmental sensor data is processed to detect anomalies. An autoencoder is a type of neural network where it is fed as input. By training the GRU-AE using historical data, the occurrence of respiratory disease, data that did not fit the model is identified by a greater reconstruction error. All reconstruction errors exceeding a threshold-based anomaly detection optimised threshold are raised. The results from the GRU-AE show that the model is able to raise alerts when such predictions deviated from the expected pattern. A change in the environment can result in occurrences of respiratory disease within 1–7 days, meaning that there is a period of time where the negative effect of respiratory diseases, such as Porcine Reproductive and Respiratory Syndrome (PRRS), a common and destructive disease endemic to pigs, can be detected.

Keywords: environmental sensors; GRU; pig; health; disease

Early detection of disease in animals is fundamental to maintaining a high level of animal health. If a disease is predicted, the sooner it can be dealt with, the better the outcome for both the animal and the farm, and increasing the number of early detections, such as porcine reproductive and respiratory syndrome (PRRS), some of the most common types of disease found in pigs, is highly valuable to be able to predict occurrences of disease at a point in time where respiratory disease prevalence is low.

Recent advances in the contraction of respiratory diseases, in particular, the identification of the respiratory disease risk factors discussed by researchers, such as temperature, and humidity—are now able to be monitored

sing inexpensive electronic sensors. Such data can be en environmental conditions and respiratory disease. sly means storing data that predominantly describes less frequent. The large amounts of normal data ality. Hence, it is better to model normality and then

nd evaluate a methodology for the early warning in growing pigs, based on environmental factors. ning (gated recurrent units (GRU)), learns how to o occurrences of respiratory disease within a batch of e used to “score” a window of sensor data according anomaly detector that was optimised using particle rmine whether to raise an alert for scores which are conditions that do not fit in line with the data it was

at there is no need for hand-crafted features, as is es, meaning that raw sensor data can be used with holds are set, as the detection of anomalous events requirement for an in-depth understanding of the nd pig health, as this is handled implicitly within the etection solution, rather than, for example, predicting ore, the method is able to perform on data consisting ormance. However, the more cases of disease there the easier it is to fine-tune and validate the method. ntial of a long short-term memory (LSTM)-based ally detection, showing the methodology has good ictable [4]. This implementation used LSTM cells, (ANN) structure, in order to handle the temporal ed to determine anomalies. However, the results of ed distinctly either in precision or recall (defined in es made in previous work, such as that by Malhotra

r GRU cells
ptimise an anomaly detector that determines whether
uated to demonstrate a more balanced anomaly

as follows. Firstly, Section 2 gives context for the n of the data and the preprocessing that was applied, the models were constructed. Section 5 outlines the by the discussion in Section 6 and the conclusions

egun to shift away from relying only on classical ncorporate machine learning (ML) methodologies, ndom Forest Classifiers, into the approach to data

or example, to detect anomalies in the production erving a range of prediction windows for the purpose

of an early warning system [5]. Other research has models) for the prediction of avian influenza prevalence performance over support vector regression [6].

Classification methods have been used to see implementation of one-class classifiers (OCCs). SVM, though there are many potential methods which offer high sensitivity but a comparatively poor specificity due to imbalance in the datasets that these methodologies are not always suitable, particularly if the true negatives are few. Cabral et al. compared a number of common methods for one-class classification (FBDOCC) of health data.

There are also approaches to anomaly detection and correlation analysis [12]. Clustering uses distance from a cluster centroid exceeds an acceptable value. However, methods falling into this category are typically of low quality of analysis and speed due to the innate complexity of analysis. Correlation analysis uses similarity metrics to determine if new data points are similar to existing ones.

Anomaly detection can be used in health applications to detect deviations from normality. A common methodology that has been shown to outperform classical statistical models, and ANOVA regression [14]. In one simulation, a network was able to correctly trigger an alarm within an average of 10 days for the model and 11–12 days for the others. Anomaly detection is often used in health applications where there are not necessarily certain conditions that define normality. For example, when detecting anomalies, it is not possible to have knowledge of all of the different conditions that define normality. It is therefore necessary to understand what normality is.

In health applications where sensors are used to monitor health, there is a need to focus not only on a high detection rate, but also on a low false alarm rate. If the false alarm rate is high, alerts from this system will not be trusted. Sensitivity to be detected in near real time [18] in order to provide timely intervention. This creates a substantial limitation as the system is resource intensive.

Often, these approaches require a transformation of the data into a useful feature representation [19–21]. However, deep learning can skip this step. Raw data can be passed to a deep neural network which is handled intrinsically by the network [16]. However, deep learning can function effectively [22]. This is particularly valuable as the network is deep enough to model the data, all data points are used.

A relatively small amount of work in agriculture has used deep learning, which mostly focuses on image classification. This demonstrates how deep learning methods, used in other domains, can be used for time-series analysis for the purpose of anomaly detection.

as applied ensemble learning methods (combined
 alence using meteorological data, showing improved

separate normality from abnormality through the
 M-based methods are very common in this approach,
 h can be applied. Models of this type usually have
 ficity [8], which is to be expected given the natural
 ies are commonly used for. However, these metrics
 negatives drastically outweigh the number of false
 mon OCCs with the proposal of features boundary
 f heart disease [8].

tion outside of classification, such as clustering [11]
 tance metrics to determine if a new point's distance
 alue: if this is the case, it is considered anomalous.
 pically unable to strike a reasonable balance between
 characteristics of time-series data [13]. Correlation
 w data is similar to known normal data.

pplications to detect outbreaks of disease or other
 ogy used for this task are Bayesian networks, which
 al methods, such as control charts, moving average
 ulation of an anthrax outbreak [15], the Bayesian
 ithin 2 days, in comparison to 9 days for the moving
 omaly detection is very well suited to healthcare, as
 that constitute a problem, but rather deviations from
 arrhythmia through electrocardiography (ECG), it is
 rent types, as new types can occur in the future [16].
 ality looks like, rather than what problems look like.
 ed, there are two main challenges. Firstly, there is
 , but also a low false positive rate [17], as, if this is
 Secondly, though not in all cases, anomalies ought
 provide a quick response so that interventions can
 on on which methods can be used, as some are very

tion be carried out on the data in order to generate a
 eep learning-based methods benefit by not requiring
 work without any hand-crafting of features, as this
 wever, this requires a substantial amount of data to
 luable for high-dimensional data, as, provided the
 ta can be utilised.

lture has made use of emerging technologies, such
 age analysis and remote sensing [23]. This paper
 in conjunction with evolutionary algorithms, can be
 nomaly detection.

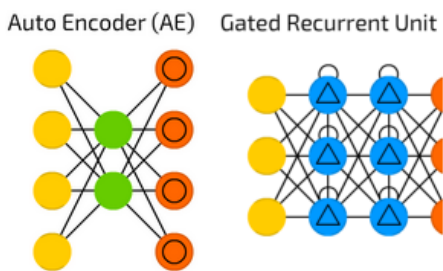


Figure 1. A visualisation of a classic autoencoder (left) and a Gated Recurrent Unit (right) [24]. These two autoencoder (AE) capable of utilising temporality.

Autoencoders are artificial neural networks (ANNs) (Figure 1). Given an input tensor N , an encoding M is trained to minimise the loss between N and N' . The attempted reconstruction of M to N' constitute performance in assisting anomaly detection, often using it as the input to an OCC [25–27], though some to detect anomalous data [28].

3. Experimental Design

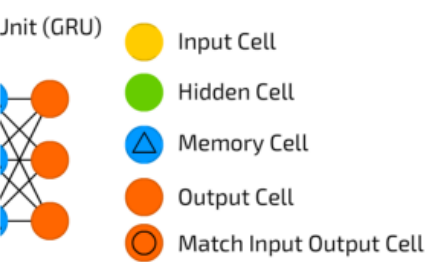
3.1. Data Collection

Data was collected for growing pigs from a varied multidisciplinary project. Each farm was represented where a batch consists of varying population sizes for varying lengths of time (14–145 days). The starting weight was 17.5 to 27.6 kg. The number of pigs per batch affected the results on a per-batch basis. Respiratory disease prevalence was divided into three levels: Individual Pig Care [29]. A pig could be either scored or treated for each level of severity. Health data regarding respiratory disease was collected by a trained person. They made observations that showed symptoms of certain diseases and clinical signs that fell. Given the nature of the system, pigs were classified as showing symptoms as such in the previous day.

The environmental data was collected using Geotag sensors placed at the centre of the room in which the batch was located. The sensors measured temperature (a sensor sensitive to $\pm 0.05^\circ\text{C}$), relative humidity (a sensor sensitive to $\pm 0.05\%$), and CO_2 concentration (with a range of reading, operating temperature range from 0°C to 40°C was sampled every minute).

3.2. Data Preprocessing

Outliers were determined to be anything outside the range of the data. Missing data induced both from the sensors and from the data were filled using linear interpolation. This was done to remove short-term variance, meaning that, on average, the data was reasonable. The results of this preprocessing of sensors



er (left) and recurrent neural network consisting of
 o methodologies are what we combined to create an
 y.

ANNs) that are trained to reconstruct input patterns
 M is found which is used to create N' . The network
 $'$. The transformation of N to M is the encoder, and
 utes the decoder. Autoencoders have shown good
 n through learning the feature representation M and
 me papers have made use of the reconstruction error

ariety of farms across Europe as part of an EU-funded
 nted by a number of batches of pigs (3–16 batches),
 es (40–1294 pigs), and the batches were monitored
 rting weight of the pigs was variable, ranging from
 cted by respiratory disease was recorded on a daily
 d into four levels of severity, as defined by Zoetis's
 ored (identified as diseased but no treatment given)
 egarding the number of pigs showing symptoms of
 manually counted the number of pigs in each batch
 classified it by the level of severity into which the
 gs cannot be identified individually each day, so the
 ptoms can be new cases or pigs that were identified

General Alert [30] sensors, which were suspended in
 ted. The sensors recorded the environmental features
 onitored temperature (a sensor range of $-50\text{ }^{\circ}\text{C}$ to
 sitivity of $\pm 2\%$ and operating temperature range of
 a range from 0 ppm to 5000 ppm ± 30 ppm or $\pm 3\%$
 $^{\circ}\text{C}$ to $50\text{ }^{\circ}\text{C}$, and humidity from 0% to 95%), which

side $\mu \pm 2\sigma$ per variable per batch and were removed
 the removal of outliers and due to hardware issues
 s deemed acceptable, as the environment has low
 e assumption of a linear change between two values
 f sensor data are shown in Figure 2. The three sensor

readings—temperature, humidity, and CO2—constituted the data collection. The four severity levels were summed and used to calculate the prevalence of respiratory disease on a given day, rather than looking at individual

$$\text{prevalence} = \frac{\text{Number of pig}}{\text{Total}}$$

This was done because respiratory diseases rarely occur in isolation. In most cases they are treated before the disease develops. There were very few cases of severe or irrecoverable respiratory disease. In the aggregation of data, the level of prevalence was still low (Figure 3). The sensor data was merged with the data from the other countries by duplicating the values for the full 24 h period.

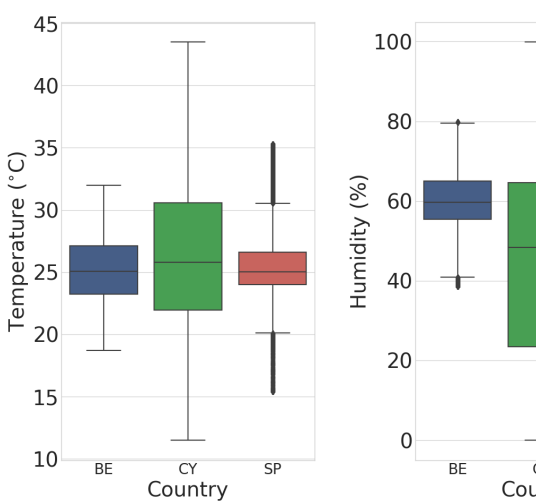


Figure 2. A boxplot describing the temperature, humidity, and CO2 levels included in the data collection—Belgium (BE), Cyprus (CY), and Spain (SP).

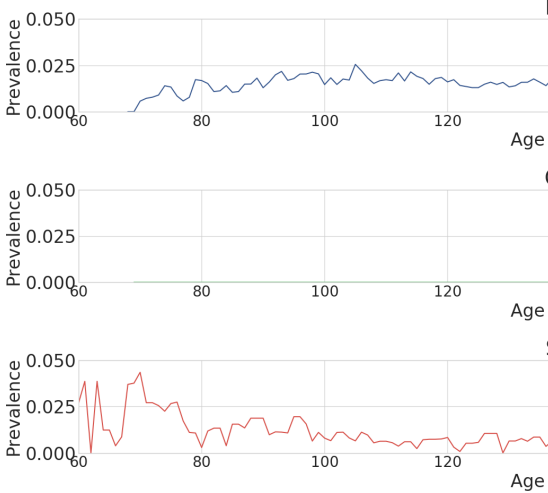
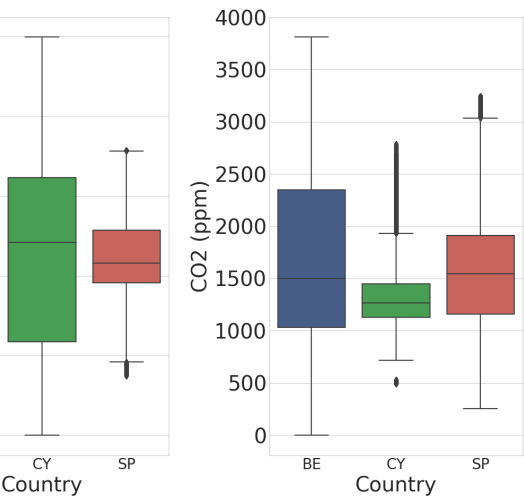


Figure 3. A graph showing the average prevalence of respiratory disease over age for growing pigs in Belgium (BE), Cyprus (CY), and Spain (SP).

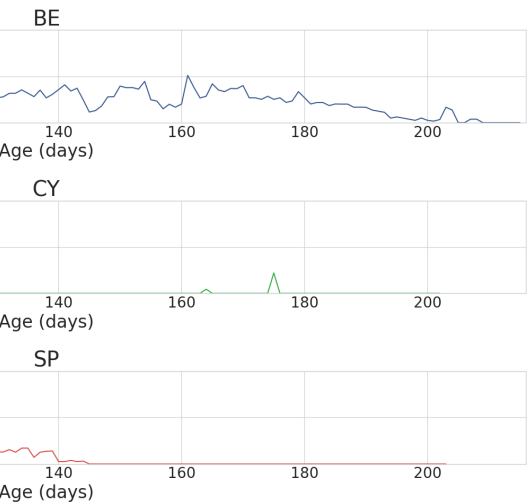
stitute the inputs to the models outlined in Section 4. calculate the proportion of pigs showing symptoms looking at one specific level of severity (Equation (1)).

pigs showing any symptoms (1)
total number of pigs

rarely exceed the classification of light or mild, as e develops into more severe symptoms. Therefore, le respiratory disease across all farms. Even with the till very low, and it decreased even further over time daily health data by upscaling the resolution of the eriod. All input was normalised to between 0 and 1.



umidity, and CO₂ sensor data for the three countries yprus (CY), and Spain (SP)—after preprocessing.



ce of respiratory disease against age (in days) in all (CY), and Spain (SP) after preprocessing.

3.2.1. Separation of Assumed Healthy and Assume

As this paper investigates the effects of environment on the existence of a lag period between a change in environment will not immediately result in an outbreak were both evaluated [32]; greater windows were not of data available for training. This lag period was u . Using this window, the data was labelled as “on unhealthy”. Data from any point where the proportion to u days after the disease prevalence decreases to 0 was labelled “assumed healthy”. All data at this point was sliding window.

3.2.2. Splitting Data into Train, Validation, and Test

It was necessary to split data into training and test sets was used to achieve this (Figure 4). Our method, (GRU-R) model used for comparison, detailed in Section 3.1. GRU network and a threshold-based anomaly detector required two sets of data for the full workflow. H_t , H_v were used for the anomaly detector. H_t was used for to validate it. O_e was used to optimise an anomaly detector was tested on to attain an evaluation metric.

O_t was a single, independent farm batch, O_e was H_v contained only the assumed healthy data of an only the assumed healthy data from all the remaining.

Each of the four datasets could be represented minutes of data in a farm batch, and each X_n contained CO_2 . This matrix was broken down into frames using timepoint (1 min) per step, creating $T \in \mathbb{R}^{m \times w \times 3}$, where

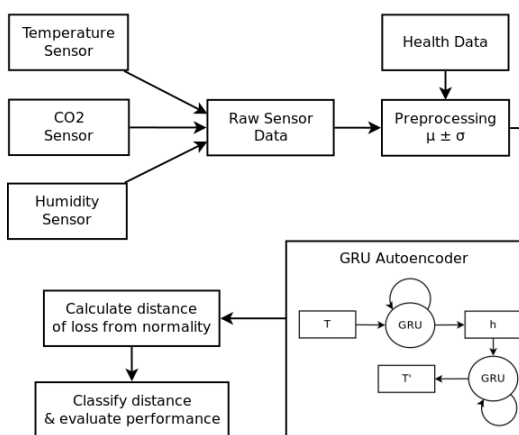


Figure 4. An overview of the entire processing workflow for anomaly detection. Temperature, CO_2 , and humidity sensors provide data. This was then broken down into four folds—used for the anomaly detection). Each of these folds is processed at 30 min. Each window of 30 min was processed and a reconstruction was produced. The distance between this reconstruction and the window is anomalous or not, using a threshold-swarm optimisation.

ed Unhealthy Data

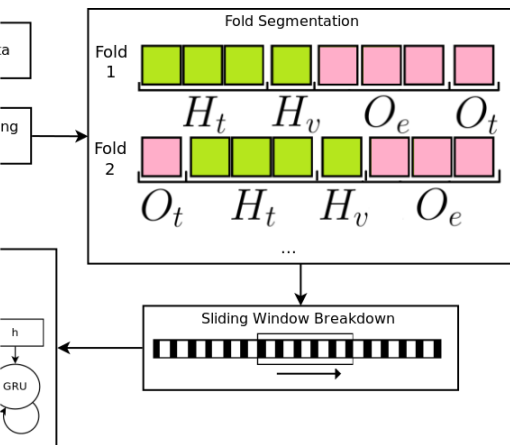
environment on health, it was necessary to account for environment and a change in health, as a change in outbreak of disease [32]. Lag periods of 14 and 21 days were not used, as this would greatly reduce the amount of data as referred to as the assumed unhealthy window, one of two sets: “assumed healthy” and “assumed unhealthy”. A portion of pigs showing symptoms was greater than 0.0 was labelled “assumed unhealthy”; all other data points were broken down into 30 min “frames” using a

st Sets

for testing sets. Leave-one-batch-out cross-validation was used, detailed in Section 4.1, and the GRU-regression

Section 4.3.3, were each composed of two parts: a detector. Both the GRU network and anomaly detector H_t, H_v were used for the GRU network, and O_e and O_t for the training of GRU networks and H_v was used for the training of the anomaly detector via PSO and O_t was the set the anomaly detector. There was no overlap between any of these sets. O_e was composed of three randomly chosen batches, another independent farm batch, and H_t contained training farm batches.

It was represented as a matrix $X \in \mathbb{R}^{n \times 3}$, where n is the number of frames. It contains sensor data for temperature, humidity, and air quality. Using a sliding window of size w , which moved one frame at a time, where m was the number of frames in the set.



The overall workflow, from the raw sensor data to anomaly detection, involved several steps. Raw sensor data were preprocessed and merged with daily health data. The data was then divided into two sets: “assumed healthy” and “assumed unhealthy”. The data was broken down using a sliding window sized by the GRU-AE, and a reconstruction error was calculated. The reconstruction error and normality was used to detect whether the data was healthy or unhealthy. The GRU-based anomaly detector optimised using particle

3.2.3. Preliminary Experiments Utilising Batch Normalisation

Batch normalisation is a deep learning technique that makes neural networks more robust by performing a per layer normalisation. The meaning of batch in this context differs from the traditional batch in the manuscript. In this context, batches are the subsets of data used for training the network being trained with the subsequent back-propagation. For anomaly detection, the effect of this technique is not necessarily positive. Based on preliminary experiments, it was found that batch normalisation allows a model to deal with inter-class variance by its mean and variance [33], meaning that the model generalises better than seen before. The decrease in performance when using batch normalisation is a requirement that the model be highly sensitive to slight changes in the final models.

4. Time-Series Early Warning Methods

4.1. GRU-Autoencoder

The implementation of an autoencoder model consists of two parts: an encoder and a decoder—that were adapted to the traditional ANN structure (Figure 1), which does not have a decoder. The encoder learned a fixed-size representation h of an input sequence, measured by using it to attempt a reconstruction of the input sequence. To minimise the reconstruction error, calculated using the input sequence T was reversed when calculating loss, as it creates similar results to the performance in sequence-to-sequence networks [34].

H_t was used to train the GRU-AE using mini-batches. The dataset contained only “assumed healthy” data, which resulted in a lower reconstruction error for this class of data than for the anomalous data. Hyper-parameters, hidden size, number of layers, learning rate, and momentum were selected using grid search, a method that evaluates the reconstruction error for each fold’s H_v were selected. To store information the GRU-AE can store for each time series, a large amount of memory for this task, the hidden size was chosen to be expected to be fairly small. The number of recurrent layers was chosen to create deep models and understanding of the time series. The mini-batch size is how many sequences are used before back-propagating the loss. This is typically described as a regularisation technique used to reduce the likelihood of overfitting in artificial neural networks. The learning rate and momentum were chosen to be small. They determine how drastically weights are updated at each iteration. Hyper-parameters were then used to calculate reconstruction errors.

Once all reconstruction errors were calculated, a threshold was determined to determine whether each reconstruction error of a given frame e was anomalous. This was estimated by calculating the Mahalanobis distance between the reconstruction error of a given frame e and the distribution of the reconstruction errors R , where $\Sigma = \text{COV}(e, R)$. This was then smoothed using a Gaussian kernel. If the smoothed distance was above some threshold (described in Section 4.1.1), then it was considered an anomalous event.

ormalisation

ique that has shown to be very effective at training
malization process on input data. Please note that
the farm batches used throughout the rest of the
ets of the training data that are fed in blocks to the
k-propagation-based error correction. In anomaly
ssarily desirable and, in this particular application,
o be detrimental to performance. This was because
ternal covariate shift by normalising each mini-batch
model is more capable of handling data it has never
using batch normalisation showed that there is a
slight variations in the data, and so it was not used

odel (Figure 4) consisted of two multi-layer GRU
ere trained simultaneously. This is instead of the
not intrinsically account for temporality in data. The
an input T . The quality of this representation was
of T using the decoder. The model was optimised
using binary cross-entropy loss, between T and T' .
s short-term dependencies that result in an increased
34].

i-batch RMSprop gradient descent [35]. This dataset
h resulted in the GRU-AE optimising towards a
ta and a higher reconstruction error for all other
of layers, mini-batch size, dropout, learning rate,
, and the parameters which resulted in the lowest
lected. The hidden size represents the amount of
series it is given. As there was no requirement for a
size representation identified by the grid search was
rrent layers represents the capability of the GRU to
series. These are expensive in terms of computation
sequences are passed forward through the network
desired to be as large as possible to speed up training,
eters and the amount of memory available. Dropout
likelihood of over-fitting and is common in standard
omentum are also used in standard artificial neural
s are changed during back-propagation. The selected
onstruction errors for all frames in O_e and O_t .
ed, the threshold-based anomaly detector was used
of a frame was to be considered anomalous or not.
is distance (Equation (2)) between the reconstruction
reconstruction errors for all frames of H_v , defined as
d using locally weighted smoothing (LOWESS) [36].
shold τ (the optimisation of which is described in
us frame.

$$d = \sqrt{(e - \bar{R})^T}$$

4.1.1. PSO-Optimised Anomaly Detection

Once all distances were calculated for all frames, an anomaly detector, capable of determining which PSO [37]—a metaheuristic, which uses concepts of swarms that produce a near-optimal solution. Consider the distance in a sequence $\vec{D} = \{d_0, d_1, d_2, \dots, d_n\}$, such that each element is the reconstruction error of a frame at index i from R . A threshold was set to determine the validity of an alert was assessed by determining if the error was within an acceptable forecasting window starting at the time an alert had been raised for this increase in prevalence, threshold was positive. Furthermore, consider the vector $\vec{F} = \{f_0, f_1, \dots, f_n\}$ of animals showing symptoms of respiratory disease at any frame $i : f_{i-1} = 0 \wedge f_i > 0$ for which there is no previous occurrence.

The parameters τ , α , and β of the detector were optimised. The particles were optimised for each fold independently. The particles were then tested on O_t for a final performance evaluation. The fitness function had three dimensions representing τ , α , and β . The fitness function was designed to maximise both Matthews correlation coefficient at the same time, minimising β . α and β were not part of the fitness function.

$$F = 2 \cdot MC$$

$$MCC = \frac{TP \times TN}{\sqrt{(TP + FP)(FP + FN)}}$$

MCC is a measure designed to evaluate the performance of a classifier robust in problems with high class imbalance, such as medical classification, -1 for perfect misclassification, and 1 for perfect classification. F-measure is commonly used in information/document retrieval problems, however, it does not include true negative rate. In document retrieval, as they are not important to the user, correctly classifying negatives (true negatives) is an important metric.

It was desirable to attain a high α , as this corresponds to the time disease is raised, whereas β needs to be minimised, as it represents the time between occurrences. For example, given the values $\alpha = 7$ and $\beta = 1$, we know that occurrences of respiratory disease will appear every 7 days. If the values $\alpha = 3$ and $\beta = 5$, we know that when an occurrence of respiratory disease will appear within 3–8 days, a new occurrence will appear. As we consider MCC to be the most important component of the fitness function in Equation 3.

4.2. Other Metrics Used for Evaluation

Specificity, a commonly used evaluation metric, measures the proportion of a large number of negative examples (the typical scenario in medical specificity, we have used positive predictive value (also known as precision) of brevity, we identify this metric as precision through Equation (6)), which we identify as recall, Equation (6)), which we identify as recall.

$$\bar{R})^T \Sigma^{-1} (e - \bar{R}) \quad (2)$$

es, the next step was to optimise the threshold-based
h frames should be considered anomalous, using
f swarm intelligence to find a set of hyper-parameters
the vector of Mahalanobis distances of all frames
each d_i is the Mahalanobis distance between the
. An alert was defined as any $d_i : d_{i-1} < \tau \wedge d_i \geq \tau$.
ining whether disease prevalence increased from 0
g at $i + \alpha$ and ending at $\alpha + \beta$. If so, and if no other
e, the alert was considered a true positive, else a false
 $f_0, f_1, f_2, \dots, f_n$ such that each f_i contains the number
ase at frame i . A false negative was determined to be
o alert within $i - \alpha$ and $i - \beta$.

r were the parameters to be optimised using PSO.
ndently in O_e , and the best values for each parameter
evaluation. Each particle was randomly initialised
The fitness function (Equation (3)) used by the PSO
lation coefficient MCC (Equation (4)) and α whilst,
ormalised between 0 and 1 when entered into the

$$CC + \alpha - \beta \quad (3)$$

$$\frac{TN - FP \times FN}{(TN + FP)(TN + FN)} \quad (4)$$

predictions of binary classifiers, which is especially
, such as our dataset. MCC scores 1 for perfect
nd 0 for equivalent to random classification. The
cument retrieval, as it does not fall prey to the same
atives in the calculation. This is not an issue within
that particular task; however, in a detection system,
an integral objective of the system.

esponds to how far in advance the alert of oncoming
d, as this represents the period of time the alert is for.
we know that when an alert is raised, it is expected
ear within 7–8 days. Given a different detector for
an alert is raised, it is expected that occurrences of
, a larger window and, therefore, more uncertainty.
ponent of our fitness function, it has a larger weight

tric, can produce misleading scores in datasets with
al scenario in anomaly detection tasks). Instead of
(also known as precision, Equation (5)). For the sake
oughout this paper. We also made use of sensitivity
ntify as recall throughout this paper. We did not use

precision or recall for the optimisation of the hyper-parameters. We then evaluate the final performance of each method.

$$P = \frac{tp}{tp + fp}$$

$$R = \frac{tp}{tp + fn}$$

4.3. Methods Included for Comparison

All methods used were tested on identical test data. To ensure statistical significance, a Wilcoxon test was conducted to compare the three comparison methods and the GRU-AE. These results were adjusted using Holm's method [38].

4.3.1. Luminol

In order to provide a baseline for comparison of time-series anomaly detection packages, Luminol [39] was used for time-series anomaly detection, as well as methods which require feature extraction, such as principal component analysis (PCA). This uses a bitmap representation of the data using symbolic aggregate approximation (SAX) [40]—to detect anomalies [41]. This is implemented using a sliding window and a leading window, which slide across the data. The distance between the two bitmaps is calculated for each window. The calculated anomaly scores are then used to threshold the data as an anomaly detector in the same way that the GRU-AE was used to detect environmental variables into a univariate time series.

4.3.2. Time-Series Regression with Autoregression Integrated Moving Average (ARIMA)

Autoregression integrated moving average (ARIMA) is a common time-series analysis in statistical analysis. The parameters are p , the order of the autoregression; d , the degree of differencing; and q , the order of the moving average. A grid search on H_t , optimising for the lowest Akaike Information Criterion (AIC) for comparing the goodness of fit of statistical models. Once the optimal parameters were found, H_v was used to predict the next 30-minute windows, given the previous 180 minutes of data. The predicted window and ground truth were compared to determine an alert for the window $i + \alpha$ to $i + \alpha + \beta$. Values for α and β were determined and, finally, tested on O_t . Similar to the Luminol method, the three environmental variables were transformed into a univariate time series.

4.3.3. Time-Series Regression with GRU Network (GRU-AE)

To provide context for how the GRU-AE performs, a GRU network was used for regression of the values of the environmental sensors. By training the GRU network, it learned to predict environmental variables for cases of respiratory disease. The model was trained to predict values for the windows used for the GRU-AE. Similar to the ARIMA model, values beyond some threshold (τ) were considered an anomaly.

r-parameters of the detector. They were only used to

$$\frac{tp}{tp + fp} \quad (5)$$

$$\frac{tp}{tp + fn} \quad (6)$$

test sets O_t . Once all methods for comparison were compared the precision, recall, and MCC between these results were corrected for multiple observations

of performance, LinkedIn's open source, univariate [39], was used. This library represents out-of-the-box univariate data to be univariate, often achieved using principal component analysis (PCA)—a method for the discretisation of time-series data implemented using two concatenated windows, a lagged version of the time series. Each window is then converted into SAX subwords based on the frequency of each SAX subword. The anomaly score for the leading component is calculated and represents the anomaly score for the leading component. The U-AE is assessed. In order to transform the three time series, the first principal component was taken.

4.2.2. Integrated Moving Average (ARIMA)

(ARIMA) is a frequently used model for univariate time series. The parameters for this model (p , the order of autoregressive model; d , the order of the moving average model) were found using Akaike information criteria (AIC). This is a metric used to compare models, though it is not a measure of goodness of fit. The model was used to fit the model. This was then used to forecast future values in O_e . Windows where the mean square error was beyond some threshold (τ) were considered as anomalies. For τ , α , and β were optimised using grid search method, the first principal component was taken to be a univariate time series.

4.2.3. (GRU-R)

GRU-R performs with regards to an alternative deep learning regression network. This network was trained to predict future values using this regression network on only healthy data H_t , under conditions which do not result in occurrences of anomalies. Predict windows of 30 minutes, the same size as the ARIMA model, windows where the mean square error was beyond an alert for the window $i + \alpha$ to $i + \alpha + \beta$. Similar to

the configuration laid out in Section 4.1, these parameters were used to train the model using O_e , and finally evaluated on O_t .

5. Results

For the purpose of exploration, the GRU-AE's parameters were searched using a grid search with a fixed $\beta = 6$ with O_v used as the test set. The results showed a steady and sustained decrease in the total number of alerts as the threshold τ increased, until the point where the acceptable forecast window extended beyond the maximum range of the acceptable forecast window. Beyond this point, an increased portion of alerts that were unverifiable in the maximum range of the acceptable forecast window were generated, leading to an increase in the total number of alerts. To reduce the search space of the PSO, and, therefore, the computational cost, the search space was also restricted by stipulating the following ranges: $0 \leq \tau \leq 20$, $1 \leq \alpha \leq 12$, and $1 \leq \beta \leq 12$. This was found through preliminary testing. β was limited to 6, as values greater than 6 were deemed to be too large.

As outlined in Section 4.1, grid search was used to find the optimal parameters. The grid search converged to the same set of parameters for all runs. In the search space, we report in Table 1 the most frequently identified parameters. The hidden size was very low. This is because under the assumption of a linear environment, and, therefore, only a small amount of non-linearity is expected. Deviations from this—frequent or drastic fluctuations—were not expected.

Table 1. The grid search-optimised hyper-parameters for the GRU-AE on multidimensional time-series data.

Hyper-Parameters

Hidden size
No of layers
Mini-batch size
Dropout
Learning rate
Momentum

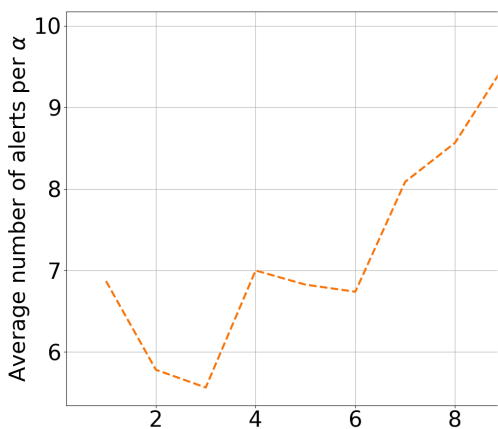


Figure 5. The average number of alerts raised, given $\beta = 7$, using the optimal threshold τ for each given α used on Mahalanobis distances produced by a GRU-AE.

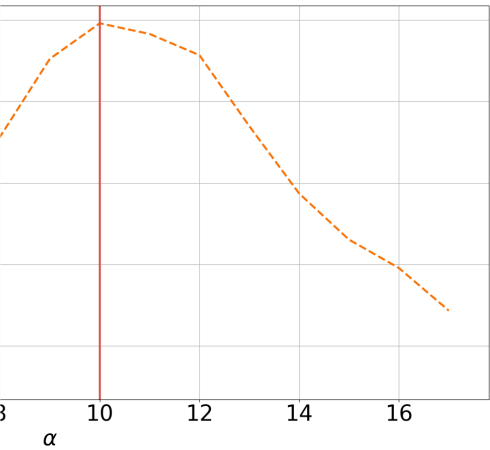
ameters were optimised using PSO, which is trained

's anomaly detector was initially optimised using a test set. For values from $\alpha = 10$ onwards, there was a sharp increase in the number of alerts raised (Figure 5). This was indicative of the model being extended beyond the data available, resulting in an increase in the number of alerts in terms of validity. This decline in alerts signified the model's performance dropping below $\alpha + \beta = 16$. This maximum endpoint was used to determine the optimal value of α . Therefore, a particle was considered invalid if $\alpha + \beta > 16$. Additionally, a particle was also considered invalid if it exceeded the range $\alpha \leq \beta \leq 6$. The limits on τ cover all feasible values; α was limited to a maximum of 6, as a range greater than

was used to find the hyper-parameters for the GRU-AE. The optimal values for the hyper-parameters in almost all batches, so, for the sake of consistency, the same values were used for all hyper-parameters. As expected, under normal and healthy circumstances there is little fluctuation in the amount of memory required. It is when there are anomalies—such as occurrences of respiratory disease can

hyper-parameters used to train a GRU-AE used on

Parameter	Value
Batch size	4
Epochs	20
Hidden layer size	1024
Learning rate	0.5
Dropout rate	1×10^{-6}
Memory	0.3



given an alert window start of α days and a size of β . The optimal values for the hyper-parameters were found using grid search on the anomaly detection GRU-based autoencoder.

Table 2 shows the batch-by-batch performance. Table 2 presents the performance of the three compared solutions. The GRU-AE achieves better results than the other two (precision, recall, and MCC). The most notable difference in methodologies is the substantial shift in performance achieved this, but to a much lower degree. In the L model, precision achieved in comparison to recall, indicating that the model is better at identifying positive changes than negative. However, in the GRU-AE model, the solutions found for precision, $\mu_P = 0.776$, $\mu_R = 0.938$, whilst not sacrificing overall performance. The substantially higher recall capacity to identify negative changes in the environment with symptoms of respiratory disease; the lower precision for negative changes in the environment result in occurring the fact that there are other factors [42] that contribute to environmental measures used in this analysis.

The last row of Table 2 contains a high-level comparison of methods across batches. We have counted the number of an incorrect score of 0 for MCC (I), correct score of 1 for precision of the C count, Luminol clearly performs the worst with a score (10–11). At the same time, GRU-AE stands out with the score of ARIMA and GRU-R, and almost four times as good as keeping the lowest count of B.

Numerically, at least, we can see that the difference in performance may be considered substantial. However, the results of the Wilcoxon test was not statistically significant for precision ($p = 0.15$). The Wilcoxon test also concluded there was no significant difference between GRU-AE, and GRU-R and GRU-AE models, in terms of recall.

Looking only at test batches where there were no alerts, the GRU model excel in determining that there are no alerts to be raised. This indicates that the GRU is capable of modeling the environment very well, as alerts are not raised when they should be. When precision was greater than zero, the ARIMA model outperformed the GRU in precision. However, this came at a substantial sacrifice in recall, likely underpowered. Both of the GRU-based models performed well, but only the GRU-AE was able to retain its high recall. The results are reported in Table 3.

ance of each of the three methodologies; each row d systems when a given batch is used for testing. her methods for all performance metrics measured difference between the proposed GRU-AE and other nce in terms of precision and recall; ARIMA has also Luminol and GRU-R models, a higher precision is t these models are more conservative in their alerts. und tend to rely on maximising recall at the expense acrificing it so much to be degrading to the model’s recall for the GRU-AE model indicates the models ironment that lead to an increased number of pigs r precision of the model could indicate that not all urrences of respiratory disease. This is backed up by ibute to respiratory disease prevalence, not just the

el quantitative summary of the performance of all mber of batches in which each method obtains either f 1 for MCC (C), or in between 0 and 1 (B). In terms rst of the methods, while the others have a similar out as having the lowest count of I, twice as low as times lower than Luminol’s, whilst simultaneously

rence in performance between Luminol and GRU-AE lts from the Wilcoxon test showed that the difference = 0.184), recall ($p = 0.059$), and MCC ($p = 0.054$). gnificant difference between the ARIMA model and rms of any of the performance metrics.

ere no events (Table 2), the deep learning models be raised, achieving extremely high precision and odelling the underlying representation of “normality” uld not be. For batches where the number of events rformed both of the GRU-based models in terms of rifice in terms of recall, indicating that the model was dels decreased in performance in terms of precision, recall. The optimised α and β values for each batch

Table 2. Table of results for the grid search-optimise integrated moving average (ARIMA) model, particle regression, and PSO-optimised threshold-based and many times the respiratory disease prevalence increase test data O_t . The final I/B/C row indicates the num

ID	Events	P	R
5	0	0.111	0.12
9	0	1.0	0.1
10	0	0.0	0.0
14	0	1.0	1.0
16	0	1.0	1.0
17	0	1.0	0.66
18	0	1.0	1.0
19	0	1.0	0.66
24	0	1.0	1.0
26	4	1.0	0.66
27	5	0.0	0.0
28	3	0.0	0.0
29	1	0.0	0.0
30	0	0.0	0.0
31	1	0.0	0.0
32	0	0.0	0.0
Avg. Events = 0		0.647	0.50
Avg. Events > 0		0.200	0.13
Avg. All		0.507	0.39
I/B/C			

ised implementation of LinkedIn’s anomaly detection library (Lu
 ticle swarm optimisation (PSO)-optimised threshold-based anom
 anomaly detection of Mahalanobis distance produced by a GRU-b
 creases from 0 within the batch, P denotes the precision, and R is t
 umber of folds that were Incorrect ($MCC = 0.0$), Between ($0.0 < M$

minol			ARIMA			GRU-R	
R	MCC	P	R	MCC	P	R	MCC
.125	0.118	1.0	1.0	1.0	1.0	1.0	1.0
0.1	0.316	1.0	1.0	1.0	1.0	1.0	1.0
0.0	0.0	1.0	1.0	1.0	0.0	0.0	0.0
1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
1.0	1.0	0.0	1.0	0.0	1.0	1.0	1.0
.667	0.816	1.0	1.0	1.0	1.0	1.0	1.0
1.0	1.0	0.0	1.0	0.0	1.0	1.0	1.0
.667	0.816	1.0	1.0	1.0	1.0	1.0	1.0
1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
.667	0.816	0.053	0.5	0.162	1.0	0.0	0.0
0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	1.0	1.0	1.0	0.0	0.0	0.0
0.0	0.0	0.0	1.0	0.0	1.0	1.0	1.0
0.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
0.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
.505	0.551	0.727	1.000	0.727	0.909	0.909	0.909
.133	0.163	0.811	0.500	0.432	0.400	0.400	0.200
.390	0.430	0.753	0.844	0.635	0.750	0.688	0.688
	7/5/4			5/1/10			5/0/11

Table 3. Table of results for the grid search-optimised library (Luminol), the grid search-optimised ARIMA detection of loss incurred from a GRU-based regression detection of the Mahalanobis distance produced by the start and length of the time window (in days), respectively, the threshold the loss/Mahalanobis distance needed to be optimised using training data O_e ; these results are for

ID	Events	Luminol		AR
		α	β	α
5	0	12	4	12
9	0	12	4	12
10	0	12	4	12
14	0	12	4	12
16	0	12	4	12
17	0	12	4	12
18	0	12	4	12
19	0	12	4	12
24	0	12	4	12
26	4	12	4	12
27	5	12	4	12
28	3	12	4	12
29	1	12	4	12
30	0	12	4	12
31	1	12	4	12
32	0	12	4	12

5.1. Case Study

The following case is presented for batch 2 methodologies applied to the problem. This batch of anomalous events in a single batch, making method to perform well on. The batch was analysed for anomaly detection (Figure 6), an ARIMA-powered (Figure 7), a GRU-R-powered, PSO-optimised anomaly detection (Figure 9). The disease prevalence in both analogue and binary form that was used for anomaly detection and the alerts this was the first principal component of the temporal GRU-R and ARIMA models, the mean square error for the GRU-autoencoder, the binary cross-entropy loss recreation of it was used. The final subplot shows the alerts, along with their alert windows.

The two deep learning-based models show similar as they both capture similar events within the environment peaks and troughs seen in the loss. The Luminol-based occurrences of respiratory disease prevalence, but false positives that it also produces. The GRU-R and in that no alerts are raised, even though the latter an anomalous points. These points rarely align with actual is largely down to not raising many alerts at all. To balance between these solutions, alerting to almost all with some false positives, though far fewer than the

sed implementation of LinkedIn's anomaly detection MA model, PSO-optimised threshold-based anomaly session, and PSO-optimised threshold-based anomaly by a GRU-based autoencoder. α and β denote the respectively, for which an alert is assessed, and τ is eded to exceed in order to raise an alert. They are e from the test data O_t .

ARIMA		GRU-R		GRU-AE	
α	β	α	β	α	β
12	1	1	6	1	1
12	1	1	6	1	2
12	1	1	6	1	2
12	1	1	6	1	4
12	1	1	6	3	1
12	1	1	6	1	5
12	1	1	6	3	5
12	1	1	6	1	5
12	1	1	6	1	6
12	1	2	6	3	6
12	4	4	6	1	6
12	1	5	6	8	6
12	2	1	6	2	6
12	1	1	6	4	6
12	1	1	6	1	6
12	1	1	6	8	5

27 exemplifies the characteristics of each of the tch was chosen as it contained the highest number ng it one of the more challenging batches for a sed by a Luminol-powered, grid search-optimised wered, grid search-optimised anomaly detection maly detection (Figure 8), and a GRU-AE-powered, he first subplot for each figure shows respiratory form; the second plots the univariate time series erts raised for it. In the case of the Luminol model, mperature, humidity, and CO₂ time series. For the rror between predicted and ground truth was used; y loss between each input window and the model's s the results of the anomaly detection for each of the

signs of consistency in the underlying technology, nvironmental data, demonstrated by the common based model correctly produces alerts to some of the ut these are severely outweighed by the number of nd ARIMA models suffer from the inverse problem r appears at first glance to produce some significant actual events, and the success of the ARIMA model . The GRU-autoencoder model, however, strikes a st all of the occurrences of respiratory disease, albeit the Luminol model.

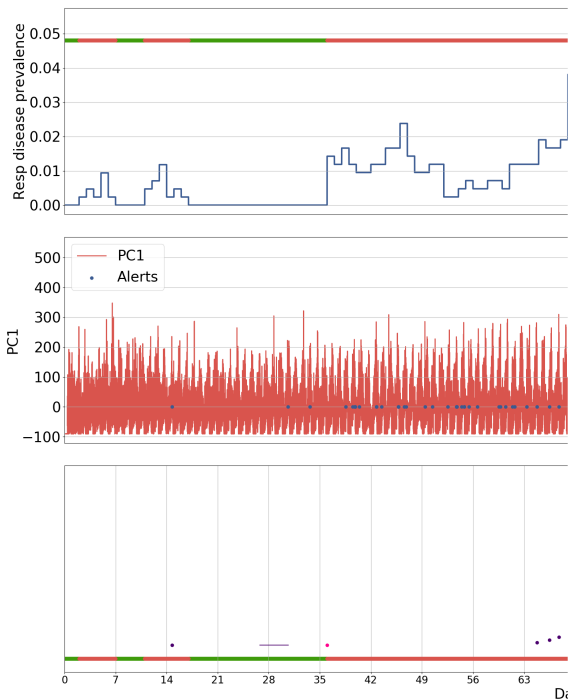


Figure 6. Results for batch 27 processed by the Lu search-optimised parameters α (the start of the win the window), in relation to the prevalence of respira

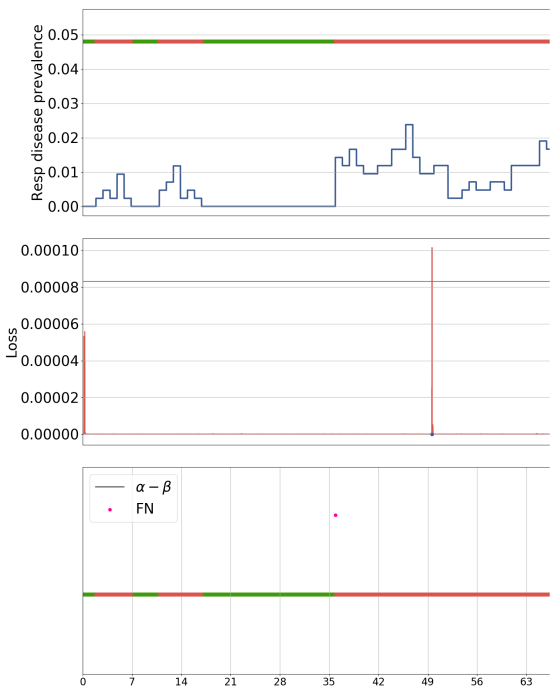
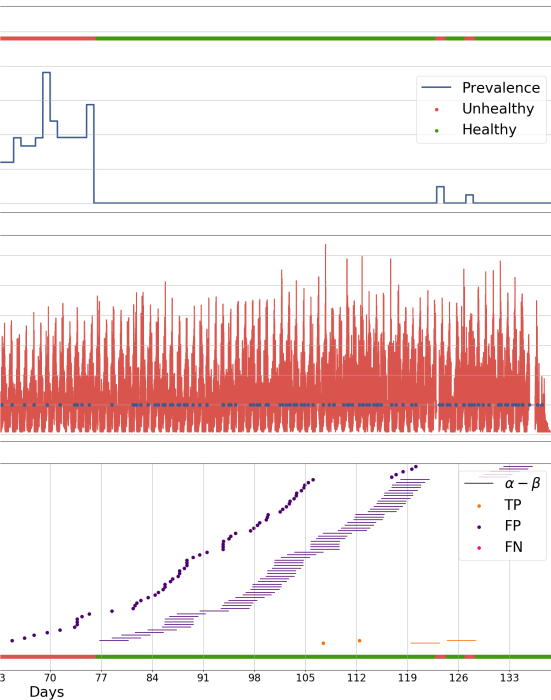
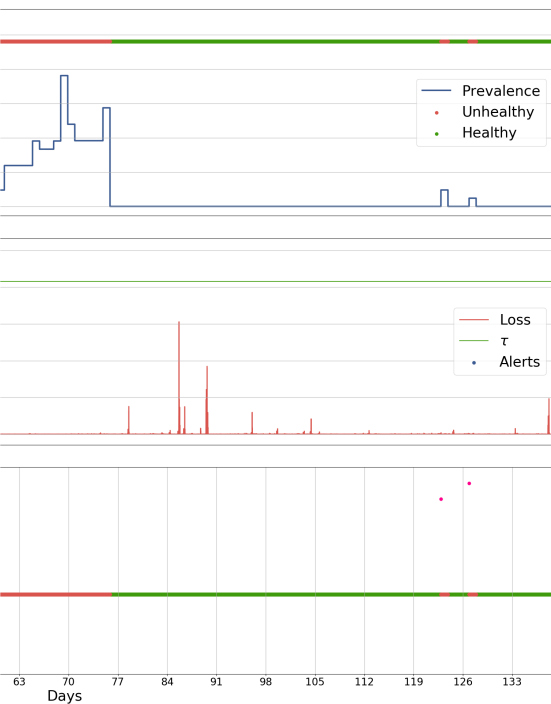


Figure 7. Results for batch 27 processed by the AR search-optimised parameters α (the start of the win window), and τ (the threshold which the loss between to be considered an alert), in relation to the prevalen



uminoI-powered anomaly detection, given its grid indow for which an alert is for) and β (the length of iratory disease within a single batch.



ARIMA-powered anomaly detection, given its grid indow for which an alert is for), β (the length of the ween predicted values and actual values must cross lence of respiratory disease within a single batch.

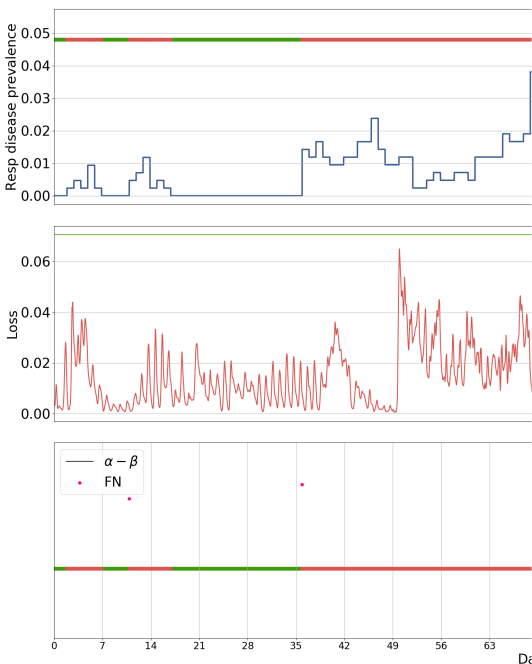


Figure 8. Results for batch 27 processed by the PSO-optimised parameters α (the start of the win the window), and τ (the threshold which the loss b cross to be considered an alert), in relation to the prev

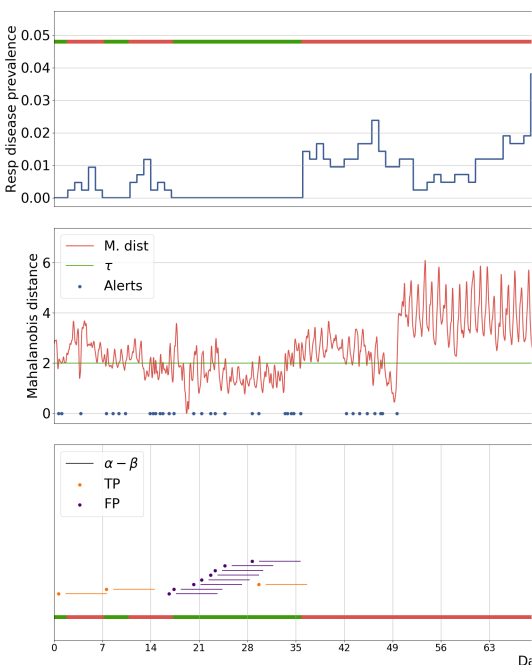
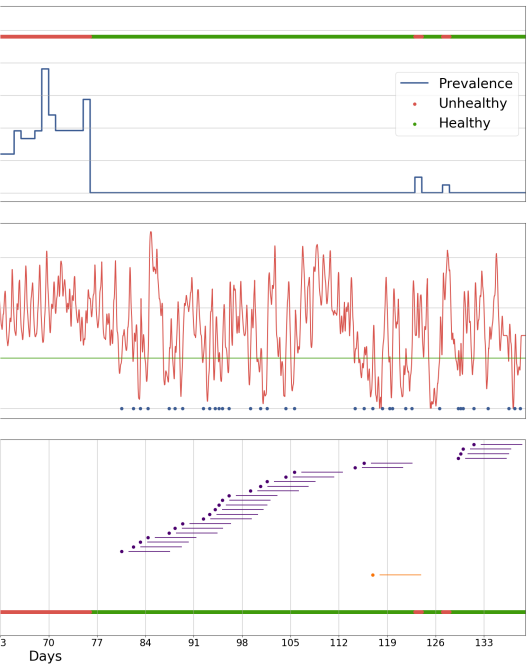


Figure 9. Results for batch 27 processed by the GRU its PSO-optimised parameters α (the start of the win window), and τ (the threshold which the loss betwe to be considered an alert), in relation to the prevalen



he GRU-R-powered anomaly detection, given its window for which an alert is for), β (the length of s between predicted values and actual values must relevance of respiratory disease within a single batch.



RU-autoencoder-powered anomaly detection, given window for which an alert is for), β (the length of the ween predicted values and actual values must cross lence of respiratory disease within a single batch.

5.2. Influence of the Number of Hidden Layers on GRU-

The number of layers used in the encoder and capable of modelling (Figure 10). Since the concept “healthy” data, the perfect environment’s loss would the model architecture should be chosen to minimise Using a small number of layers, 1–15, there was litt “healthy data”. Once the number of layers reached reached its maximum and would not reduce the sta

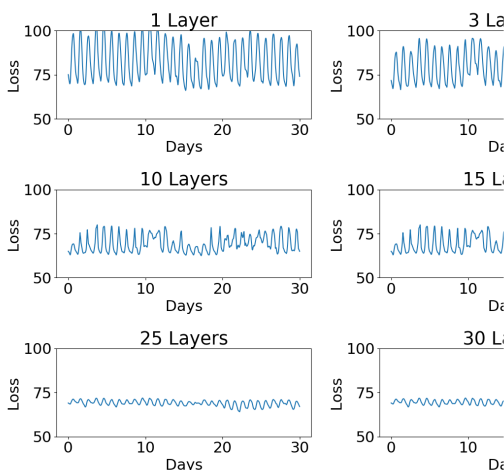


Figure 10. How different numbers of layers in a G data. The smaller the standard deviation, the better

5.3. Computation Time

Despite its lacking in performance, the Lumin and optimised in a fraction of the time required to tr model was fully trained within a period of 6 h and t 24 h. These models were run on a machine conta 44 cores), whereas the two GRU models took a we with an Nvidia GTX 980 Ti.

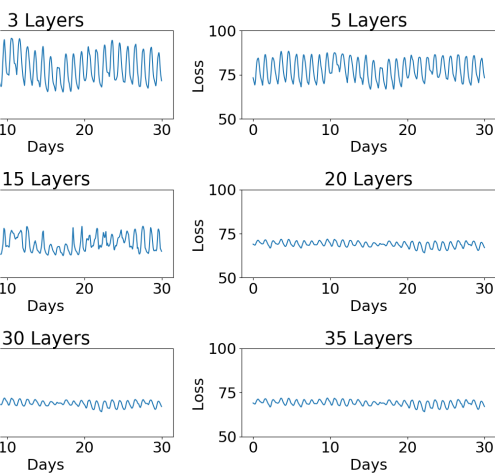
6. Discussion

This paper set out to develop and evaluate a comprising GRUs, and to apply the methodology growing pigs. This network structure was used to the temporal aspect of data within the structure of t structuring of the data. The GRU-AE specifically ou be detected, resulting in a substantially higher recall

The novelty of this work lies in the complete reas multi-sensor anomaly detection of Malhotra et al. [3 from LSTM cells, currently the most common uni the network as a whole, as GRUs are less comple comparable performance [43–46]. A PSO-optimised the output of the GRU-AE to determine if a win methodological changes, this work provides a mu

U-AE Performance

nd decoder had an effect on what the GRU-AE was
cept behind the model was that loss is reduced for
ld have a constant loss over time, therefore meaning
ise the standard deviation of the loss in healthy data.
little change in the ability of the GRU-AE to model
ed 20 for each half of the GRU-AE, the performance
standard deviation of loss any further.



GRU-AE affect the ability to model environmental
ter the model performs.

inol-based model presented (Figure 6) was trained
train the two GRU models presented. The Luminol
d the ARIMA-based model was fully trained within
ntaining two Xeon E5-2699v4 processors (totalling
week on a similarly performing machine equipped

a methodology utilising recurrent neural networks
gy to the early detection of respiratory disease in
to build an autoencoder capable of accounting for
of the network itself, rather than solely through the
outperforms the GRU-R where there were events to
call for the GRU-AE.

assessment of the idea of a recurrent autoencoder for
[3]. The primary adjustments made were switching
nit in recurrent networks, to GRUs. This benefits
plex and therefore more efficient whilst providing
ed anomaly detector was also created for processing
indow was anomalous or not. In addition to the
uch more in-depth evaluation of the performance

of the GRU-AE through the use of a case study and whilst also applying the methodology to real-world

Given the recent trends in livestock keeping, portions of livestock management and monitoring high-quality, sustainable, and safe meat production [opportunities for automation in disease detection, such to embracing what is generally referred to as precision variety of sensors. Small, relatively cheap, and high variables, such as those assessed in this study, and sensors such as cameras [48] and microphones [49]. I have the potential to lead to epidemics and therefore this reason, a high precision ought to be sacrificed if pose less danger than false negatives in this application upon this, ensuring that recall is maximised as much

The motivation for developing such a method prevalence in the data that was collected. In this work diseases. There were several reasons for this, including diseases was highest amongst this class of pigs in this that affect pigs, such as lameness and digestive disorders respiratory diseases, developing similar methodologies occurrences to validate the model.

Steps were taken to mitigate the challenges posed of respiratory disease (light, mild, severe, and irregular justified, as this methodology is for an early-warning disease or if there is not. Therefore, looking at respiratory segmented into levels of severity, makes for a simple being tackled. It should be noted that the methodology respiratory diseases in pigs.

The detection of respiratory disease following the parameters derived from literature, such that pathogen incubation (e.g., PRRS has an average incubation GRU-AE are able to raise an alert for oncoming respiratory first showing symptoms. The Luminol model does inability to find hyper-parameters that result in a stable values for α and β , as there is no repercussion for doing

Due to the renewed popularity of RNN-based exceptional performance in fields such as natural language increase the interpretability of these black-box models and future work might consider applying such methods of the risk factors for respiratory disease in growing assumption-free and works without considering distribution validate that this multi-sensor anomaly detection scales thus validating its robustness, would be valuable to

7. Conclusions

We have presented a deep learning-based model of respiratory disease in growing pigs. The proposed combined with a PSO-optimised anomaly detection anomalies that will lead to occurrences of respiratory country-specific environments. This GRU-AE was able

and concrete comparisons with alternative methods, world data.

g, there is a move towards automating significant changes in response to the ever-increasing demand for precision [47]. These changes in livestock management offer such as the one developed here. A common approach in precision livestock farming (PLF) is the introduction of a high-resolution sensors can monitor environmental conditions and also capture large quantities of other data using sensors [48]. In the case of disease detection, any false negatives or false positives can increase the disease risk of livestock systems. For this reason, if a higher recall can be achieved, as false positives are reduced, the GRU-AE presented in this paper acts as good as possible.

The methodology was to deal with the very low disease prevalence in the dataset, we concentrated on the detection of respiratory diseases, including the fact that the prevalence of respiratory diseases in this dataset. We have similar data for other disorders and diseases, but as their incidence is much lower than for respiratory diseases, methodologies would result in an insufficient number of

features used by the data, such as summing all classifications (if not recoverable), as described in Section 3.2. This was the case for the training system, which only needs to know if there is a respiratory disease prevalence as a whole, rather than a specific prevalence per dataset that is more focussed on the challenge of the methodology developed here can be applied to all types of

data. A change in the environmental condition fits within the dataset, but as we are not predicting oncoming disease before the incubation time of 14 days [32]). Both the GRU-R and the GRU-AE for respiratory disease 1–7 days prior to pigs in a batch. The results do not reflect this, however, this is likely due to its low specificity and a strong MCC. This causes the PSO to maximise the performance by doing so if the performance is poor regardless.

For the trained models for sequence analysis, a result of their application in natural language processing, there is a need for methods to improve the models. Work in this area has steadily grown [50,51], and the application of methods to the model to gain a better understanding of the data. Using pigs based on the GRU-AE. As this method is applied to the differences between countries, additional work to be done in the scales beyond the domain of health within agriculture, and to the wider community.

The model is capable of raising an early warning to occurrences of disease. The proposed methodology of a GRU-based autoencoder shows strong potential in its ability to detect respiratory disease in growing pigs, and it is robust to noise and is able to outperform other comparable methodologies.

in precision, recall, and overall performance in both s of disease.

Author Contributions: Conceptualization, Data Curation, Validation, Visualization, and Writing—Original Draft P Investigation, Project administration, Resources, Super Conceptualization, Methodology, Writing—Review an Methodology, Validation, Writing—Review and Editing, a

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